

Measure Preservation and Dyadic Volume Decomposition in the EABC Coordinate Space

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Abstract

We investigate an infinite dyadic decomposition of a local source volume within the four-dimensional EABC coordinate system. The construction induces a natural probability measure on the four orthogonal coordinate directions

$$(E, A, B, C).$$

We prove that the decomposition is measure preserving, convergent, and complete. The resulting weight vector

$$\left(\frac{8}{15}, \frac{4}{15}, \frac{2}{15}, \frac{1}{15}\right)$$

defines a canonical point in the positive simplex and simultaneously determines a normalized quaternionic direction on the unit three-sphere.

The decomposition therefore separates additive volume conservation from geometric direction normalization.

1 Introduction

The EABC framework is based on the four invertible residue classes modulo 12

$$E \equiv 1, \quad A \equiv 5, \quad B \equiv 7, \quad C \equiv 11 \pmod{12}.$$

These classes form the Klein four-group

$$V_4 = \{E, A, B, C\}$$

with multiplication laws

$$A^2 = B^2 = C^2 = E, \quad AB = C, \quad AC = B, \quad BC = A.$$

In the present work we consider a local source volume spanned by four orthogonal coordinate directions

$$\hat{e}, \hat{a}, \hat{b}, \hat{c}.$$

A dyadic four-phase decomposition generates a canonical hierarchy of weights associated with these directions.

2 Dyadic Four-Phase Expansion

Definition 1. *Define the EABC expansion*

$$\Xi = \sum_{k=0}^{\infty} \left(2^{-(4k+1)} \hat{e} + 2^{-(4k+2)} \hat{a} + 2^{-(4k+3)} \hat{b} + 2^{-(4k+4)} \hat{c} \right).$$

Since the EABC cycle repeats every four steps, each coordinate direction is scaled by

$$q = 2^{-4} = \frac{1}{16}.$$

The individual coordinate contributions are therefore geometric series.

3 Coordinate Weights

Lemma 1. *The four coordinate weights are*

$$\Xi_e = \frac{8}{15}, \quad \Xi_a = \frac{4}{15}, \quad \Xi_b = \frac{2}{15}, \quad \Xi_c = \frac{1}{15}.$$

Proof. For the E direction,

$$\Xi_e = \sum_{k=0}^{\infty} 2^{-(4k+1)} = \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{1}{16} \right)^k.$$

Using the geometric-series identity,

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}, \quad |q| < 1,$$

gives

$$\Xi_e = \frac{1}{2} \cdot \frac{1}{1 - \frac{1}{16}} = \frac{8}{15}.$$

The remaining coordinates follow analogously:

$$\Xi_a = \frac{4}{15}, \quad \Xi_b = \frac{2}{15}, \quad \Xi_c = \frac{1}{15}.$$

□

4 Measure Preservation

Theorem 1 (EABC Measure Preservation). *The dyadic four-phase decomposition induces a probability measure*

$$\mu = (\Xi_e, \Xi_a, \Xi_b, \Xi_c)$$

satisfying

$$\mu(E) + \mu(A) + \mu(B) + \mu(C) = 1.$$

Proof. Using the previous lemma,

$$\frac{8}{15} + \frac{4}{15} + \frac{2}{15} + \frac{1}{15} = \frac{15}{15} = 1.$$

□

Consequently,

$$\mu = \left(\frac{8}{15}, \frac{4}{15}, \frac{2}{15}, \frac{1}{15} \right)$$

defines a probability measure on the EABC coordinate set.

5 Volume Conservation

Definition 2. *Let the local source volume be normalized by*

$$V_M = 1.$$

Corollary 1. *The dyadic decomposition preserves total volume in the additive sense:*

$$V_M = \Xi_e + \Xi_a + \Xi_b + \Xi_c.$$

Thus the decomposition neither creates nor destroys volume. No gaps and no overlaps occur with respect to the induced measure.

6 Euclidean Geometry

The corresponding EABC vector is

$$\Xi = \frac{8}{15}\hat{e} + \frac{4}{15}\hat{a} + \frac{2}{15}\hat{b} + \frac{1}{15}\hat{c}.$$

Because the coordinate axes are orthogonal,

$$\|\Xi\|_2^2 = \left(\frac{8}{15} \right)^2 + \left(\frac{4}{15} \right)^2 + \left(\frac{2}{15} \right)^2 + \left(\frac{1}{15} \right)^2.$$

Hence

$$\|\Xi\|_2^2 = \frac{64 + 16 + 4 + 1}{225} = \frac{85}{225} = \frac{17}{45}.$$

7 Quaternionic Interpretation

Identify the EABC basis with the quaternion basis

$$E \leftrightarrow 1, \quad A \leftrightarrow i, \quad B \leftrightarrow j, \quad C \leftrightarrow k.$$

Then

$$Q = \frac{8}{15} + \frac{4}{15}i + \frac{2}{15}j + \frac{1}{15}k.$$

The quaternion norm satisfies

$$|Q|^2 = \frac{17}{45}.$$

Therefore

$$\hat{Q} = \sqrt{\frac{45}{17}} Q$$

lies on the quaternionic unit sphere

$$S^3.$$

8 The Role of Fifteen

The number

$$15 = 8 + 4 + 2 + 1$$

appears naturally as the denominator of the normalized measure.

Remarkably,

$$15 = 2^4 - 1$$

is also the number of non-empty subsets of a four-element set.

Hence the denominator coincides with the complete combinatorial content of a four-phase system.

9 Conclusion

The EABC dyadic decomposition induces a canonical probability measure

$$\left(\frac{8}{15}, \frac{4}{15}, \frac{2}{15}, \frac{1}{15} \right)$$

that preserves total volume exactly.

The construction separates two distinct structures:

1. additive measure conservation,
2. geometric normalization on the quaternionic unit sphere.

This provides a mathematically consistent bridge between dyadic hierarchical decompositions, measure-preserving dynamics, and quaternionic geometry.